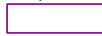
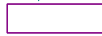


Daisy



File: Seed3-DevKit-Desktop\_1.kicad\_sch

Peripherals



File: Seed3-DevKit-Desktop\_2.kicad\_sch

Audio



File: audio.kicad\_sch

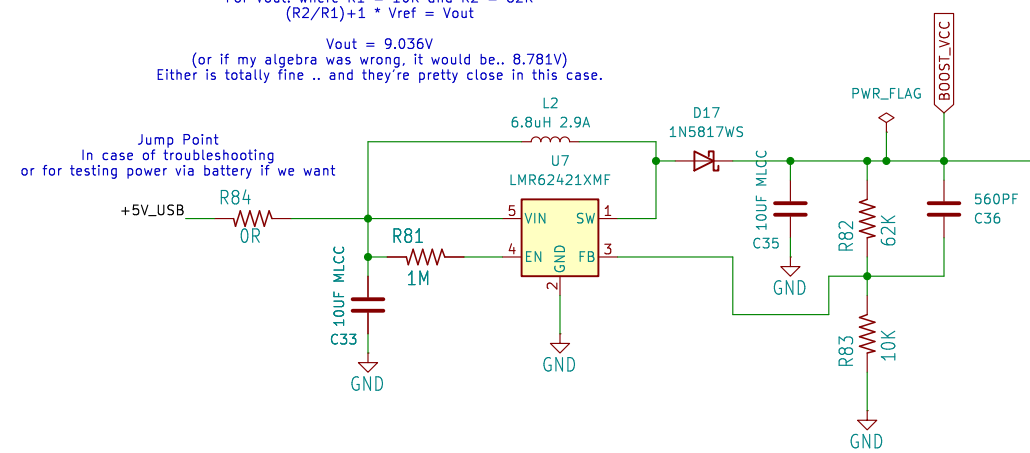
POWER

Selecting R2 with formula:  
 $R2 = (V_{out}/V_{ref} - 1) \times R1$   
Where  $V_{ref} = 1.255V$

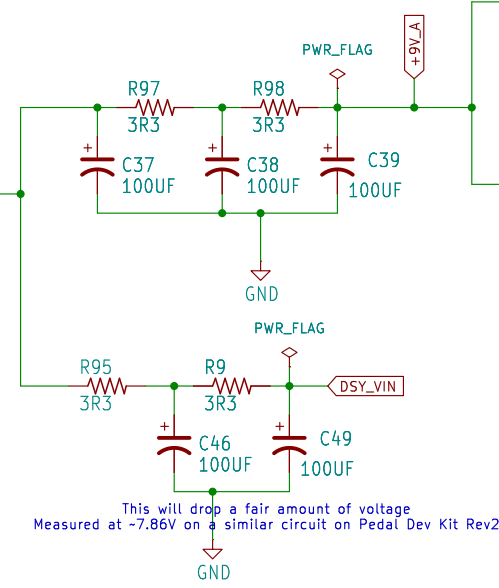
For  $V_{out}$ : where  $R1 = 10K$  and  $R2 = 62K$   
 $(R2/R1)+1 \times V_{ref} = V_{out}$

$V_{out} = 9.036V$   
(or if my algebra was wrong, it would be.. 8.781V)  
Either is totally fine .. and they're pretty close in this case.

Jump Point  
In case of troubleshooting  
or for testing power via battery if we want

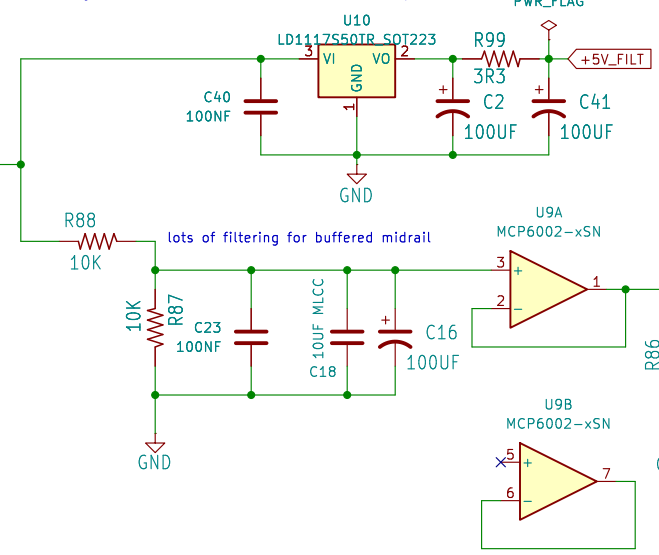


This will drop a fair amount of voltage  
Measured at -8.6V on similar circuit on Pedal Dev Kit Rev2



We can remove this, and simplify slightly if we:

- Power Headphone Amp from +3v3\_A
- Change DAC output op-amp to a rail-to-rail opamp that supports higher voltage (i.e. OPA1652)
- Power DAC output op-amp from +9V\_A
- Add LM4040 for +5V reference for CV inputs
- Use TL074 or something for AREF\_AUDIO\_BIAS buffer instead, and power at +9V



Decide if we wanna float this or GND it like usual..

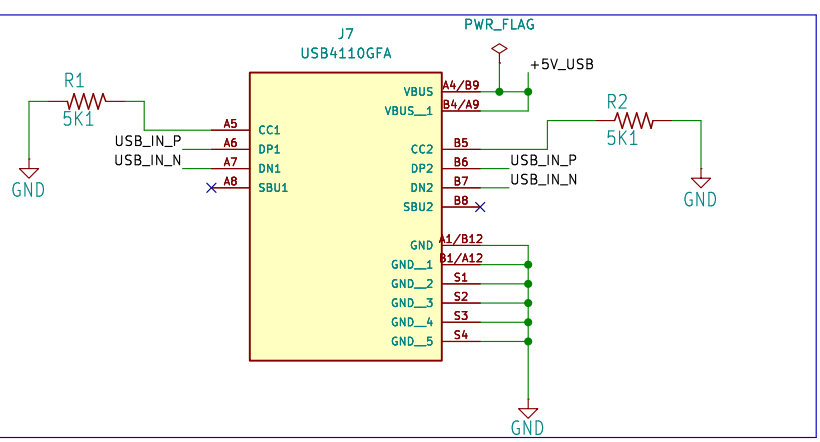
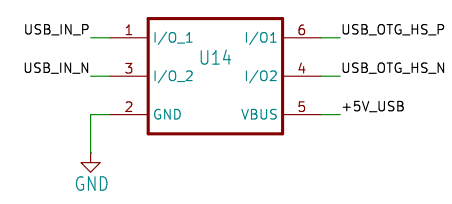


QR1

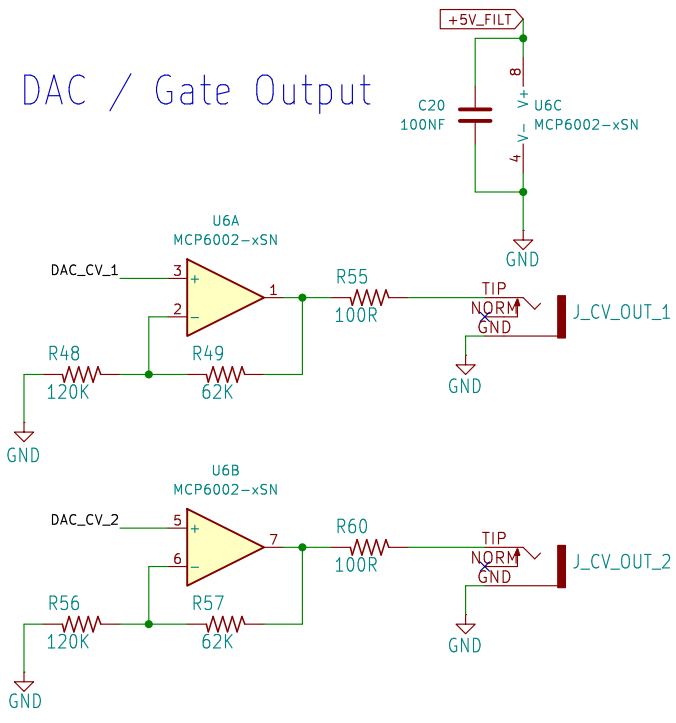


- H1 MOUNT-PAD-ROUND3.5
- H2 MOUNT-PAD-ROUND3.5
- H3 MOUNT-PAD-ROUND3.5
- H4 MOUNT-PAD-ROUND3.5

USB C

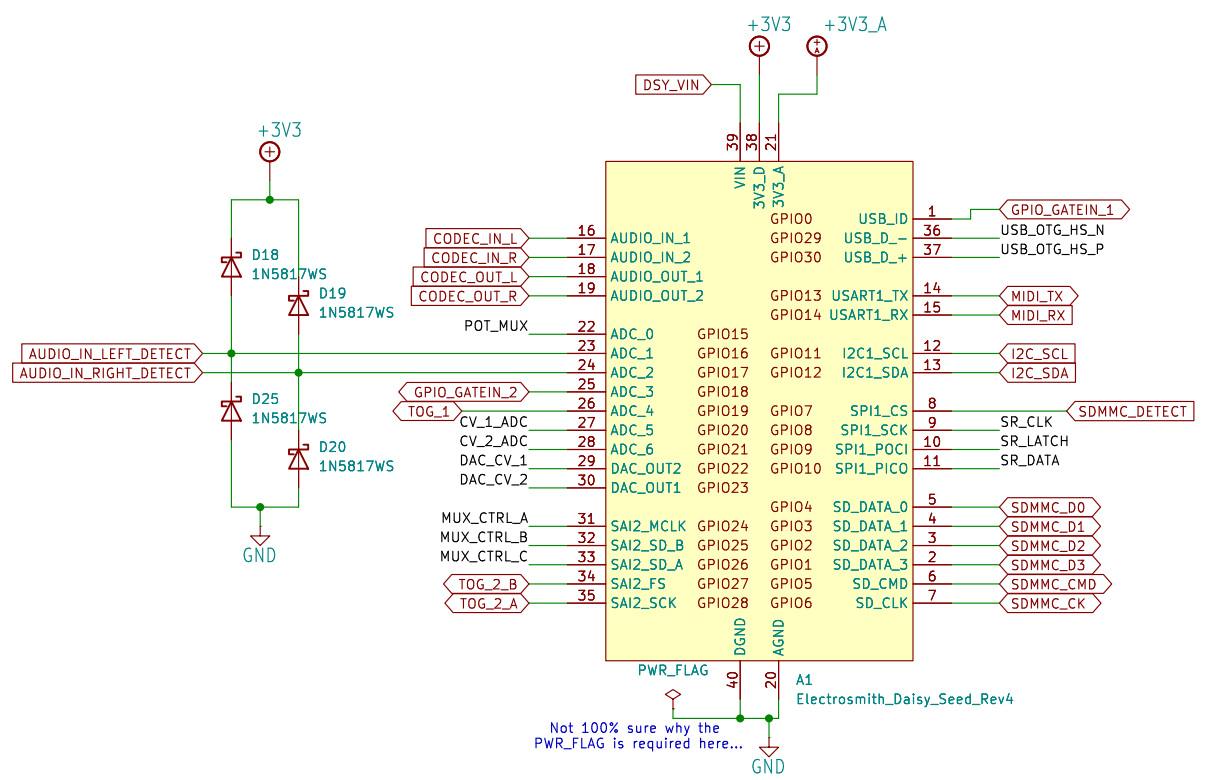
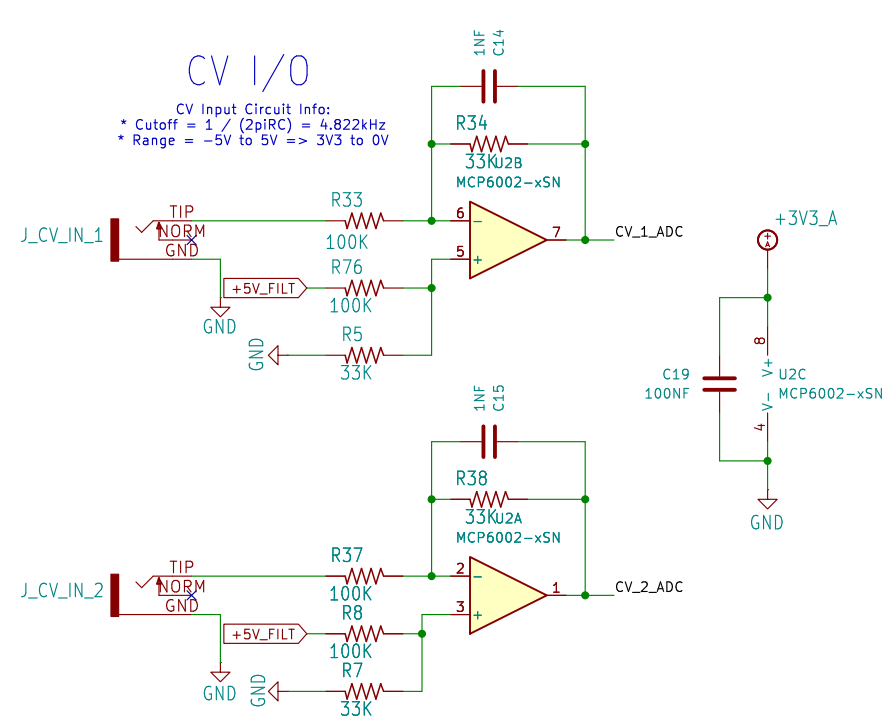


DAC / Gate Output



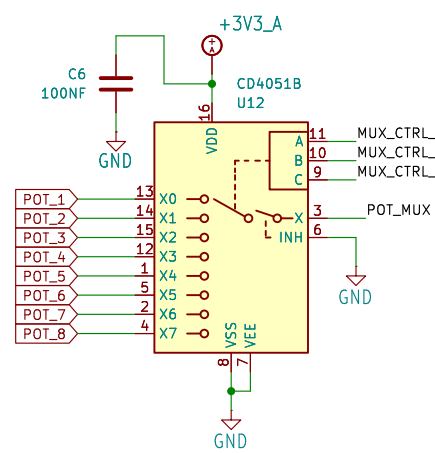
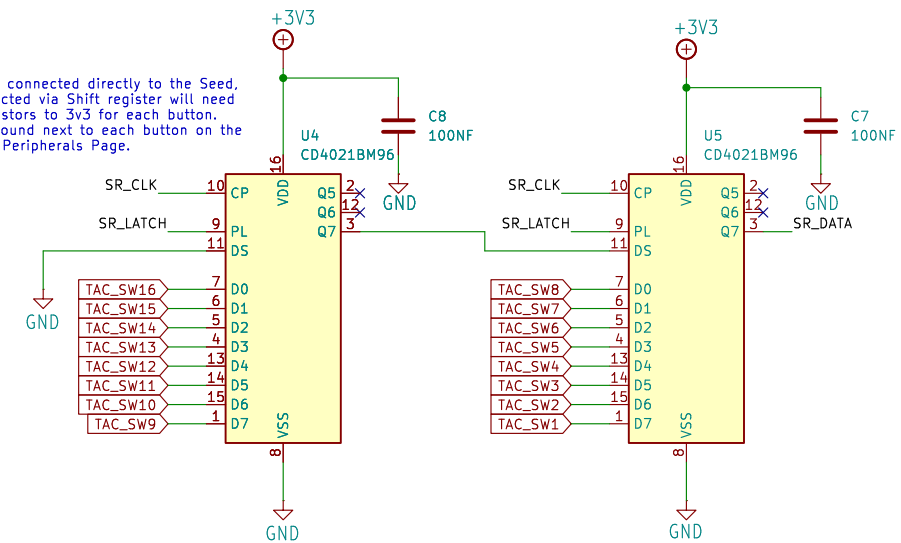
CV I/O

CV Input Circuit Info:  
\* Cutoff =  $1/(2\pi RC) = 4.822kHz$   
\* Range = -5V to 5V => 3V3 to 0V



Not 100% sure why the  
PWR\_FLAG is required here...

Unlike buttons connected directly to the Seed,  
buttons connected via Shift register will need  
pull up resistors to 3v3 for each button.  
These can be found next to each button on the  
Peripherals Page.



# LEDs

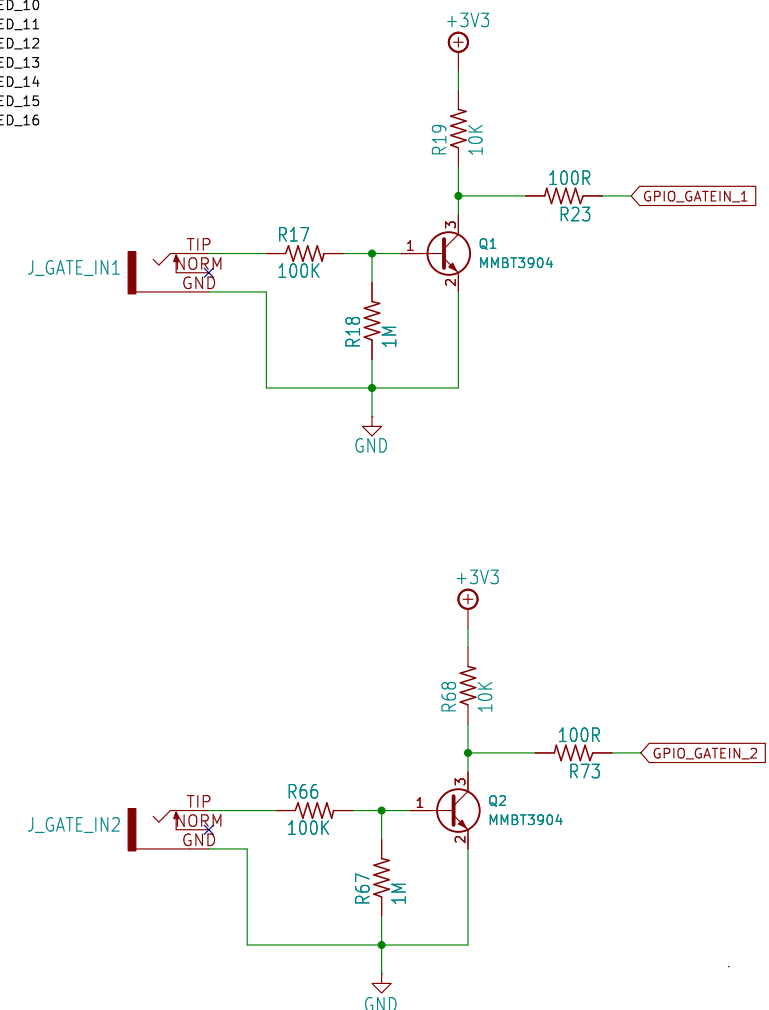
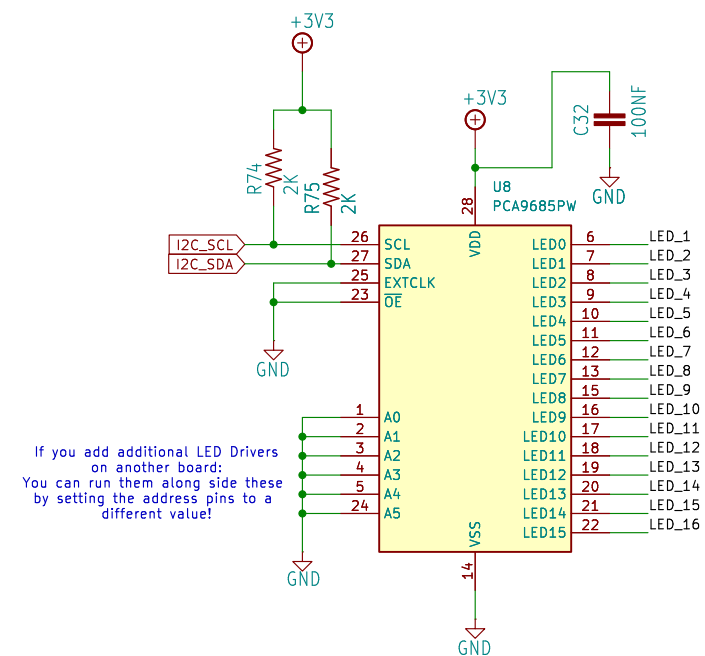
The diagram illustrates 16 individual LED connections, labeled LED\_1 through LED\_16. Each connection consists of a 2K resistor (R43, R40, R45, R69, R35, R41, R46, R70, R36, R42, R47, R71, R39, R44, R58, R72) connected in series with an LED (D1, D5, D9, D13, D2, D6, D10, D14, D3, D7, D11, D15, D4, D8, D12, D16). The LEDs are connected to ground (GND).

# MIDI

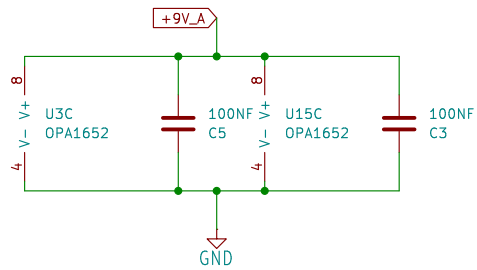
The diagrams show three different MIDI connection configurations using 74VHC111L (U11, J10, J8) monostable multivibrators.

- Top Diagram:** MIDI IN (MIDI\_IN\_4, MIDI\_IN\_5) connected to MIDI RX. The circuit includes a 3V3 supply, a 220R resistor (R4), a 100nF capacitor (C1), and a 270R resistor (R3).
- Bottom Left Diagram:** MIDI IN connected to MIDI IN\_4 and MIDI IN\_5. The circuit includes a 3V3 supply, a 100nF capacitor (C34), a 100nF capacitor (C47), and a PWR\_FLAG signal.
- Bottom Right Diagram:** MIDI RX connected to MIDI THRU. The circuit includes a 3V3 supply, a 10R resistor (R10), and a 10R resistor (R29).

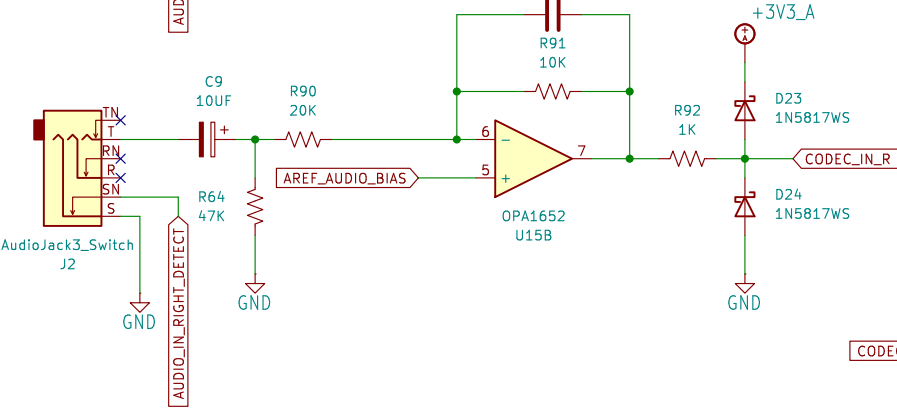
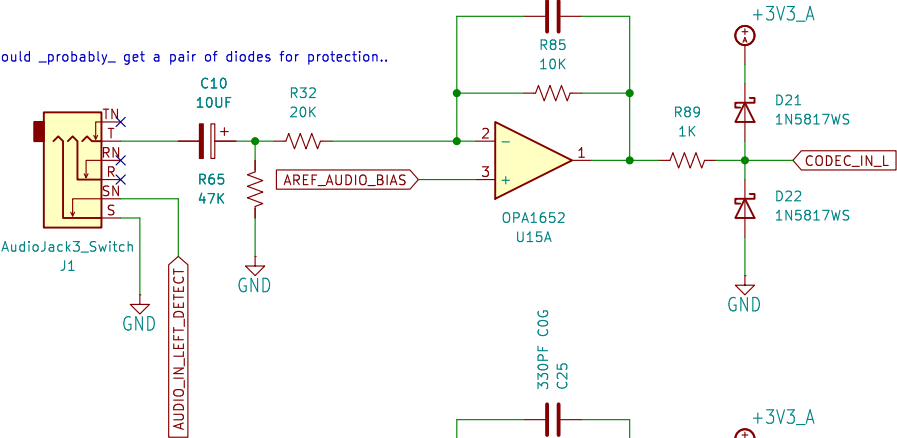
The PWR\_FLAG signal is noted as being awkward to implement here, but it is recommended for input isolation.



## AUDIO I/O



...s should \_probably\_ get a pair of diodes for protection..

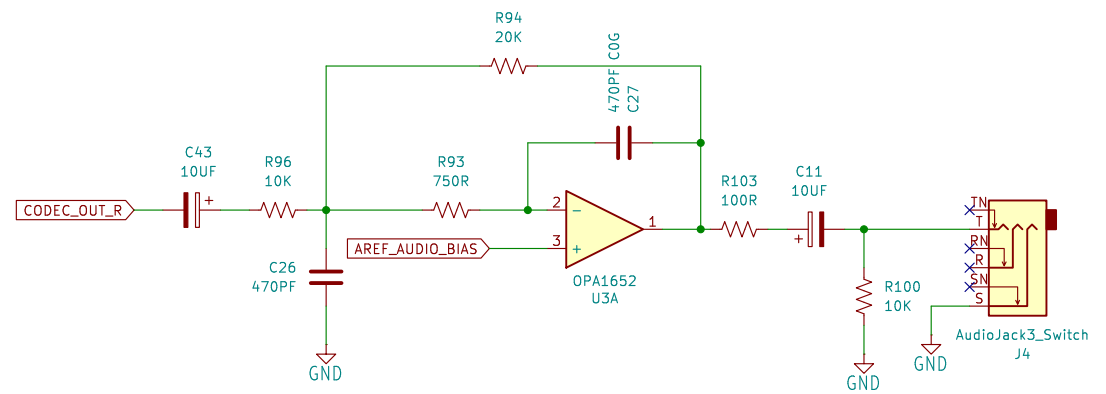


Gain is set to target +4dBu (maximum)  
obviously this does not make it a pro-line-level device that way..  
we'd need to probably have a certain amount of headroom.  
If, instead we look at this as a -10dBV device, then we have  
-14dB of headroom with these values.

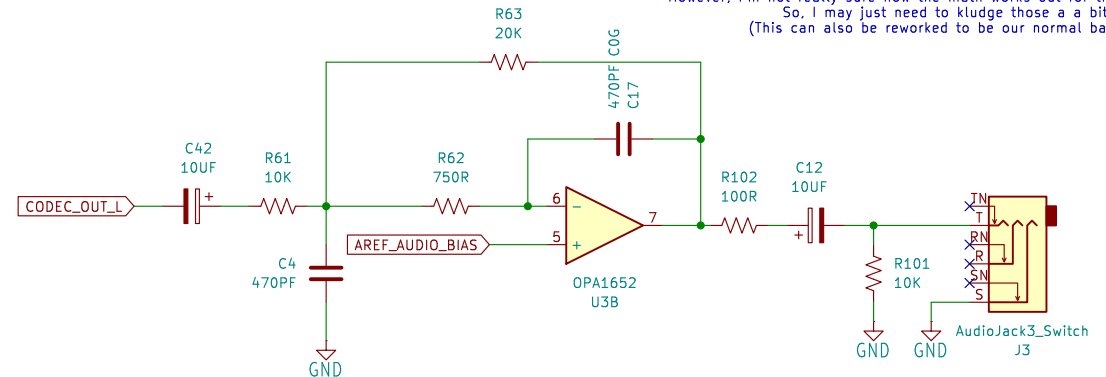
$G_{in}=0.814$  ( $R_{in}=10k$ ,  $R_{fb}=16k$ )  
 $G_{out}=1.228$  ( $R_{in}=10k$ ,  $R_{fb}=24k$ )

Alternatively we can go closer to the PCM3060's "2Vrms" reference and just do:  
 $G_{in}=0.5$ ,  $G_{out}=2$  :shrug:

For now, I've implemented the left channel of audio with less atten/gain.



I updated these to be gain of 2.  
However, I'm not really sure how the math works out for the other two values..  
So, I may just need to kludge those a bit IRL.  
(This can also be reworked to be our normal basic filter)



DECISION - Two Choices:  
1. Add LD1117 (or LP2985) or other 5V Linear Reg.  
2. Power TPA6110 from Seed's 3v3A

